Quiz #2

physics_abstract

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. A particle is located relative to the origin at the position vector $\vec{r} = -7\hat{i} + 11\hat{j}$, where distances are measured in arbounits. Which of the following statements is correct?

A. The distance from the particle to the origin is 18 arbounits.
This comes from adding the components, $7 + 11 = 18$, instead of finding the magnitude of the vector. Distance from the origin must be computed with the Pythagorean relation, not by direct addition.

B. The distance from the particle to the origin is 11 arbounits.
This mistake treats the y-component as if it were the full distance. It ignores the nonzero x-component of the position vector.

C. The distance from the particle to the origin is 7 arbounits.
This mistake treats the x-component magnitude as the total distance. It neglects the contribution from the y-component.

D. None of the above is correct.
The distance from the origin is the magnitude of the position vector, $|\vec{r}| = \sqrt{(-7)^2 + 11^2} = \sqrt{170} \approx 13.0$ arbounits. Since 13.0 is not equal to 18, 11, or 7, the only correct choice is that none of the listed numerical values is correct.

Answer: *D*

Q2. If the displacement of an object is given by $\vec{s}(t) = \left((2.1\,\mathrm{m/s^3})t^3 - (1.7\,\mathrm{m/s^2})t^2\right)\hat{i}$, which of the following statements is correct?

A. This is a constant-acceleration motion.
This assumes the acceleration does not change with time, but differentiating a cubic displacement produces a time-dependent acceleration. The presence of the t^3 term guarantees that $\vec{a}(t)$ is not constant.

B. The magnitude of the acceleration at $t = 0$ is $3.4\,\mathrm{m/s^2}$.
Differentiate the displacement twice: $\vec{v}(t) = \left(6.3t^2 - 3.4t\right)\hat{i}$ and $\vec{a}(t) = \left(12.6t - 3.4\right)\hat{i}$. At $t = 0$, the acceleration is $-3.4\hat{i}\,\mathrm{m/s^2}$, so its magnitude is $3.4\,\mathrm{m/s^2}$.

C. The magnitude of the acceleration is $1.7\,\mathrm{m/s^2}$.
This confuses the coefficient of the t^2 term in displacement with the acceleration itself. After two derivatives, that coefficient contributes $-3.4\,\mathrm{m/s^2}$, not $1.7\,\mathrm{m/s^2}$.

D. The magnitude of the acceleration at $t = 0$ is $0\,\mathrm{m/s^2}$.
This error comes from focusing only on the cubic term, whose contribution to acceleration vanishes at $t = 0$. It overlooks the quadratic term, whose second derivative gives a nonzero constant contribution.

Answer: *B*

Q3. An object takes 4.0 s to move from position $\vec{r}_A = 1.0\hat{i} - 2.0\hat{j}$ to position $\vec{r}_B = 7.0\hat{i} + 6.0\hat{j}$, where position is measured in meters. Which of the following statements is correct?

A. The path length traveled by the object is 10 m.
This uses the displacement magnitude, $\sqrt{(7-1)^2 + (6-(-2))^2} = 10\,\text{m}$, as though it were the actual distance traveled along the path. The path length depends on the trajectory and cannot be determined from just the initial and final positions.

B. The speed at $t = 0$ must be 2.5 m/s.
This comes from dividing the displacement magnitude by the total time to get $10/4 = 2.5\,\text{m/s}$, then incorrectly interpreting that as the initial speed. That calculation gives neither the initial speed nor necessarily even the average speed along the actual path.

C. None of the above is correct.
From the given data, we can determine only the displacement, $\Delta\vec{r} = (6\hat{i} + 8\hat{j})\,\text{m}$, whose magnitude is $|\Delta\vec{r}| = 10\,\text{m}$. However, the actual path length, the speed at $t = 0$, and the acceleration at $t = 0$ are not uniquely determined by just two positions and one elapsed time, so the other statements are not necessarily true.

D. The acceleration at $t = 0$ must be 0.63 m/s^2.
This assigns a specific acceleration value from insufficient information. Infinitely many possible motions can connect the same start and end positions in 4.0 s, each with different accelerations.

Answer: *C*